

Week 8 Knowledge Check — AI Spec Development & Capstone

A short retention check covering the week's core ideas: the AI Spec Development pattern, capstone discovery and the compressed PRD, compressed architecture (TCD/TMD-light), technical-logic validation and finalization, and Friday's presentations. Answer each question, then check yourself against the key at the end. Aim for concepts, not trivia — every answer maps back to something we used in the capstone.

Format: 12 questions (multiple choice + true/false). ~15 minutes. Individual or quad.

Questions

1. (MC) In the AI Spec Development pattern, the 5-prompt sequence is run how?

- A) All 5 prompts in parallel, then validate the assembled spec once at the end
- B) Each prompt validated before the next runs — a validation chain
- C) As one mega-prompt that includes every input at once
- D) Prompts 1–4 draft the spec; prompt 5 validates all four together

2. (T/F) Senior TPMs use AI Spec Development to draft the spec, not to do the thinking — judgment shapes every section and the AI accelerates the typing.

3. (MC) In the capstone, "compressed" work means:

- A) Lower-quality work produced faster
- B) Selective work — keep what's load-bearing, cut what's cosmetic
- C) Only the happy-path sections, skipping edge cases
- D) AI-generated drafts run with lighter validation

4. (MC) In the compressed PRD-light, the acceptance-criteria coverage target is:

- A) 6 happy / 0 sad / 0 weird
- B) 3 happy / 2 sad / 1 weird
- C) 2 happy / 2 sad / 2 weird
- D) 5 happy / 1 sad / 0 weird

5. (MC) When running the AI Spec sequence under time pressure, which content must be preserved verbatim rather than paraphrased by the AI?

- A) The engineering-ready summary paragraphs
- B) Trade-off costs, revisit triggers, named invariants, and customer quotes
- C) The headline and the section titles
- D) The provenance log entries

6. (T/F) In the Day-3 cross-quad spot-check, a reviewer quad that reads the AI Spec and reports "looks fine" without checking claims against the source has done its job correctly.

7. (MC) The 7-point technical-logic validation on Day 4 exists mainly to check:

- A) That each artifact is well-written on its own
- B) Whether the artifacts fit together and cohere as one set
- C) That the AI Spec matches the presentation deck word-for-word
- D) That grammar and formatting are consistent across documents

8. (T/F) Most quads pass all 7 technical-logic checks on the first pass.

9. (MC) In the cross-artifact integration pass, a good cross-reference must be:

- A) Short and informal — "see the TCD" is enough
- B) Specific (cite the Section), verified, and add value
- C) Placed only in the AI Spec, not the other documents
- D) Generated by AI to guarantee consistency

10. (MC) On the Friday presentation rubric, which dimension carries the single highest weight?

- A) Customer + problem clarity (15%)
- B) AI Spec quality (20%)

- C) Trade-off honesty (15%)
- D) Presentation craft (15%)

11. (T/F) Quads are scored on a curve — ranked against one another so the scores spread out.

12. (MC) During Q&A, when asked a specific question you cannot answer, the disciplined response is:

- A) Give a confident best guess so you don't look unprepared
- B) "Honestly, I don't know — here's what we'd do to find out"
- C) Immediately defer the question to the instructor
- D) Restate your customer + problem to fill the time

Answer Key

1. B — Each prompt validates before the next runs. The validation chain is what keeps the AI Spec honest; running all five in parallel breaks it.

2. True — "Senior TPMs use this to draft, not to think." Judgment shapes every section; AI accelerates the typing. AI Spec without validation is fiction.

3. B — Compression is selective, not lower-quality: keep what's load-bearing, cut what's cosmetic. Every word earns its place.

4. B — 3 happy / 2 sad / 1 weird. Compressed means *fewer* ACs, not weaker ones — still Given/When/Then, same 5-failure-mode discipline.

5. B — Trade-off costs, revisit triggers, named invariants, and customer quotes are preserved verbatim; everything else can be synthesized, but hallucinated claims get cut.

6. False — Watch for rubber-stamping. The reviewer should find at least one weak claim; "looks fine" without checking against the source is the failure mode.

7. B — Each artifact may be fine on its own; the 7 checks ask whether they fit together (e.g., SLO budget fits the sequence diagram, data model supports the API).

8. False — Most quads fail 2–3 checks on the first pass. That's expected — that's why the finalization day exists.

9. B — Specific (cite the Section), verified, and adds value. "See the TCD" without a section number is a sign the writer never actually verified it.

10. B — AI Spec quality at 20% is the highest-weighted dimension — the week's signature artifact carries the most weight.

11. False — Score against the rubric, not against other quads. Whole-number scores, no curve: if two quads earn 3.5, both are 3.5.

12. B — "Honestly, I don't know — here's what we'd do to find out." Curiosity, not defensiveness; a well-handled "I don't know" beats a hand-wave.